

# Confidence Intervals

## Lecture 10

# Topics in this lecture

- Central Limit Theorem
- Large-sample confidence interval for a population mean

# Central Limit Theorem

- If we draw a large enough sample from the population, then the distribution of the *sample mean* is approximately *normal*, no matter what population the sample was drawn from
- Let  $X_1, X_2, \dots, X_n$  be a simple random sample drawn from a population with mean  $\mu$  and variance  $\sigma^2$ ,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

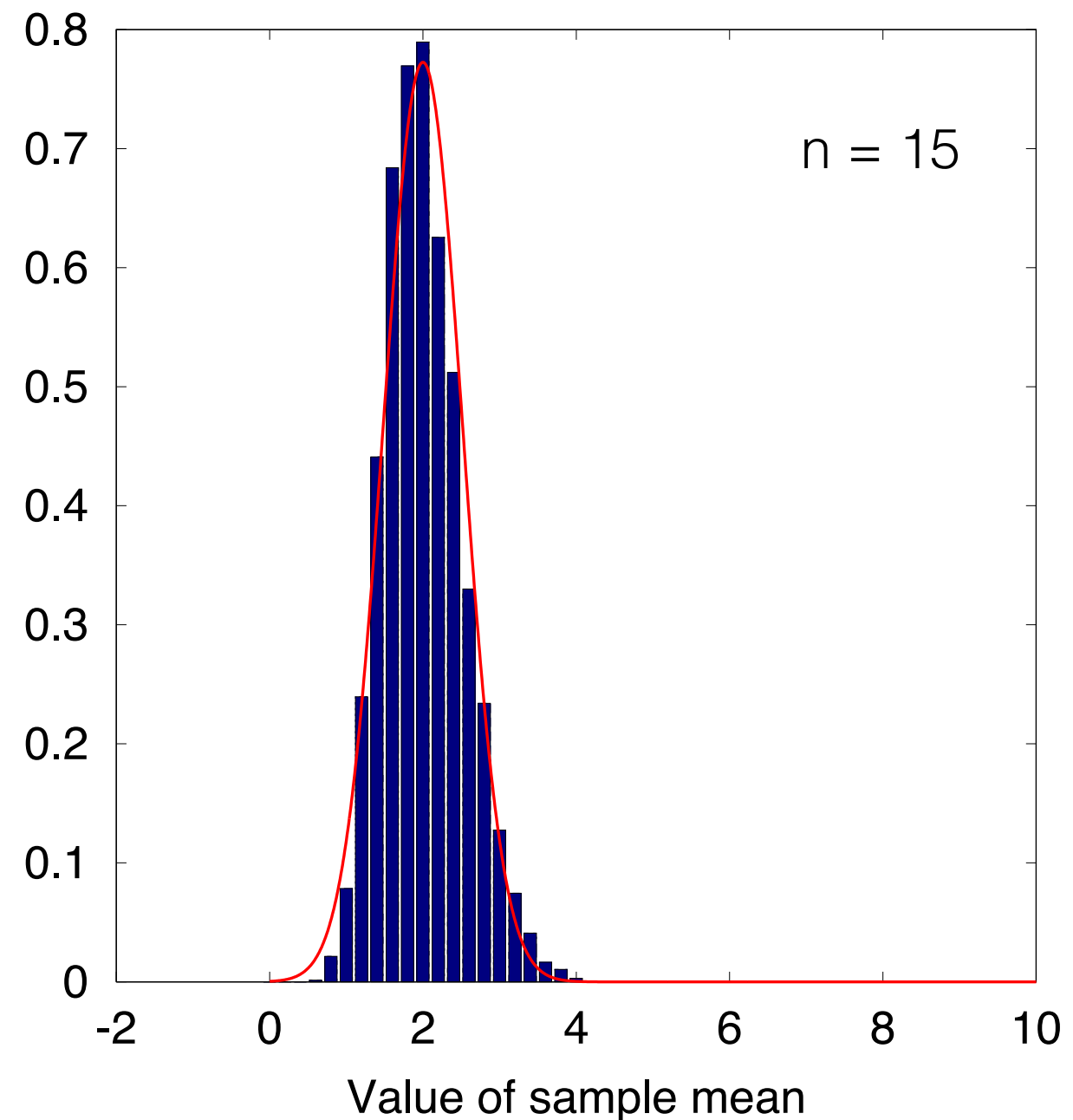
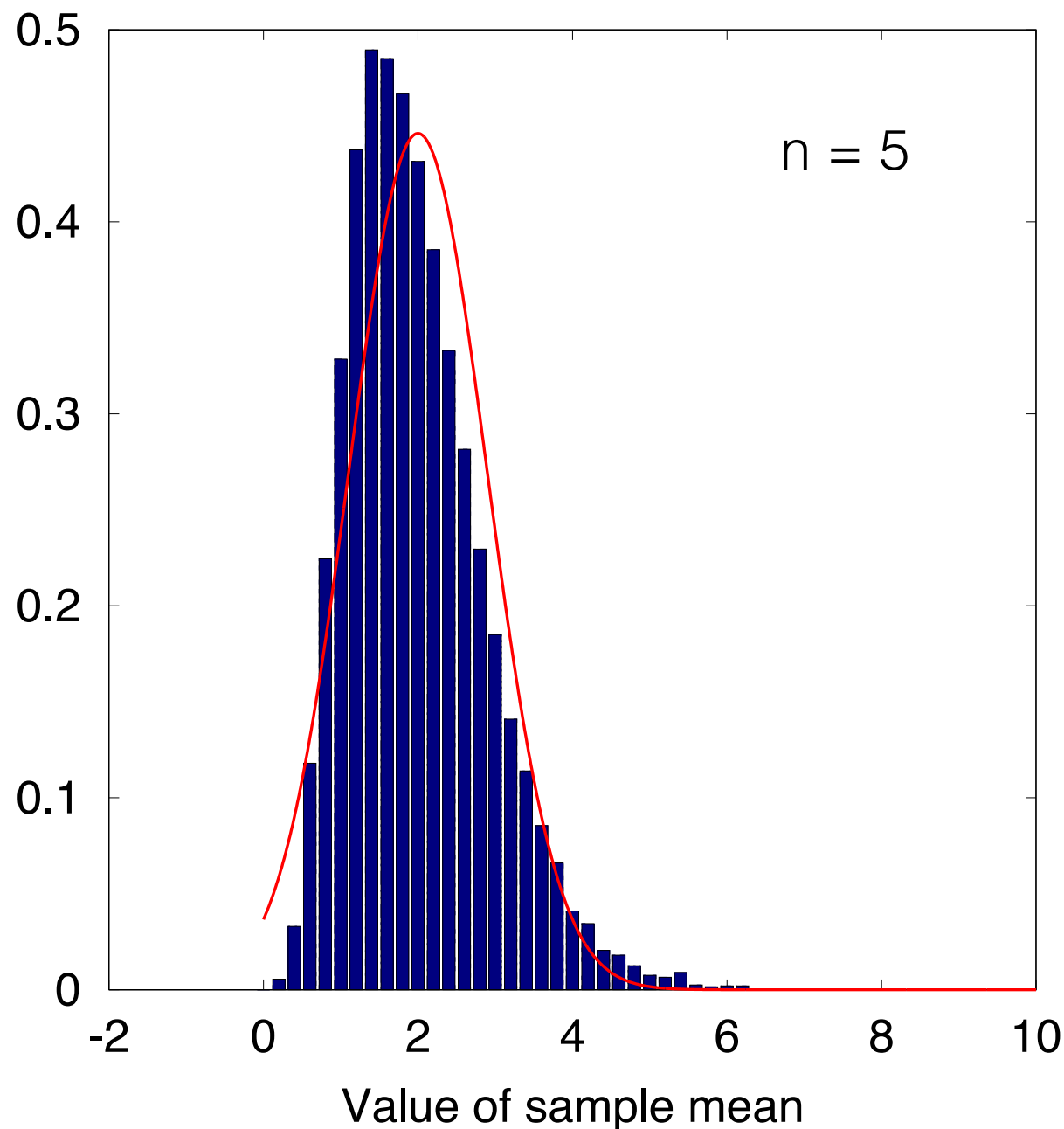
$$\bar{X} \sim \mathcal{N} \left( \mu, \frac{\sigma^2}{n} \right)$$

if  $n$  is large enough ( $n \geq 30$ )  


$$S_n = \sum_{i=1}^n X_i$$

$$S_n \sim \mathcal{N} (n\mu, n\sigma^2)$$

# Distribution of Sample Mean



Distribution of sample mean. Samples are drawn from a population with  $\text{Exp}(0.5)$

# Example 1 *z-table*

- In a large school, the mean weight of the students is 53.4 kg, and the standard deviation is 6 kg. A random sample of 50 students is drawn. What is the probability that the average weight of these students will be greater than 55 kg?

$$\mu = 53.4, \sigma = 6$$

$$P(\bar{x} > 55)$$

$$\bar{x} \sim N\left(53.4, \frac{6^2}{50}\right)$$

$$P(\bar{x} > 55) = 1 - P(\bar{x} \leq 55)$$

$$P(\bar{x} > 55) = 1 - P\left(z \leq \frac{55 - 53.4}{\sqrt{\frac{6^2}{50}}}\right)$$

$$= 1 - \Phi(1.89)$$

$$= 1 - 0.9706 = 0.0294$$

# Confidence Intervals

- Sample mean is only an estimator of the true mean. It is not exactly equal to the true mean.
- We want to assess the level of *confidence* that the true mean falls within a certain interval

# Confidence Interval for Population Mean

- Let  $X_1, X_2, \dots, X_n$  be a large ( $n \geq 30$ ) random sample from a population with mean  $\mu$  and standard deviation  $\sigma$ .

- We know that

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

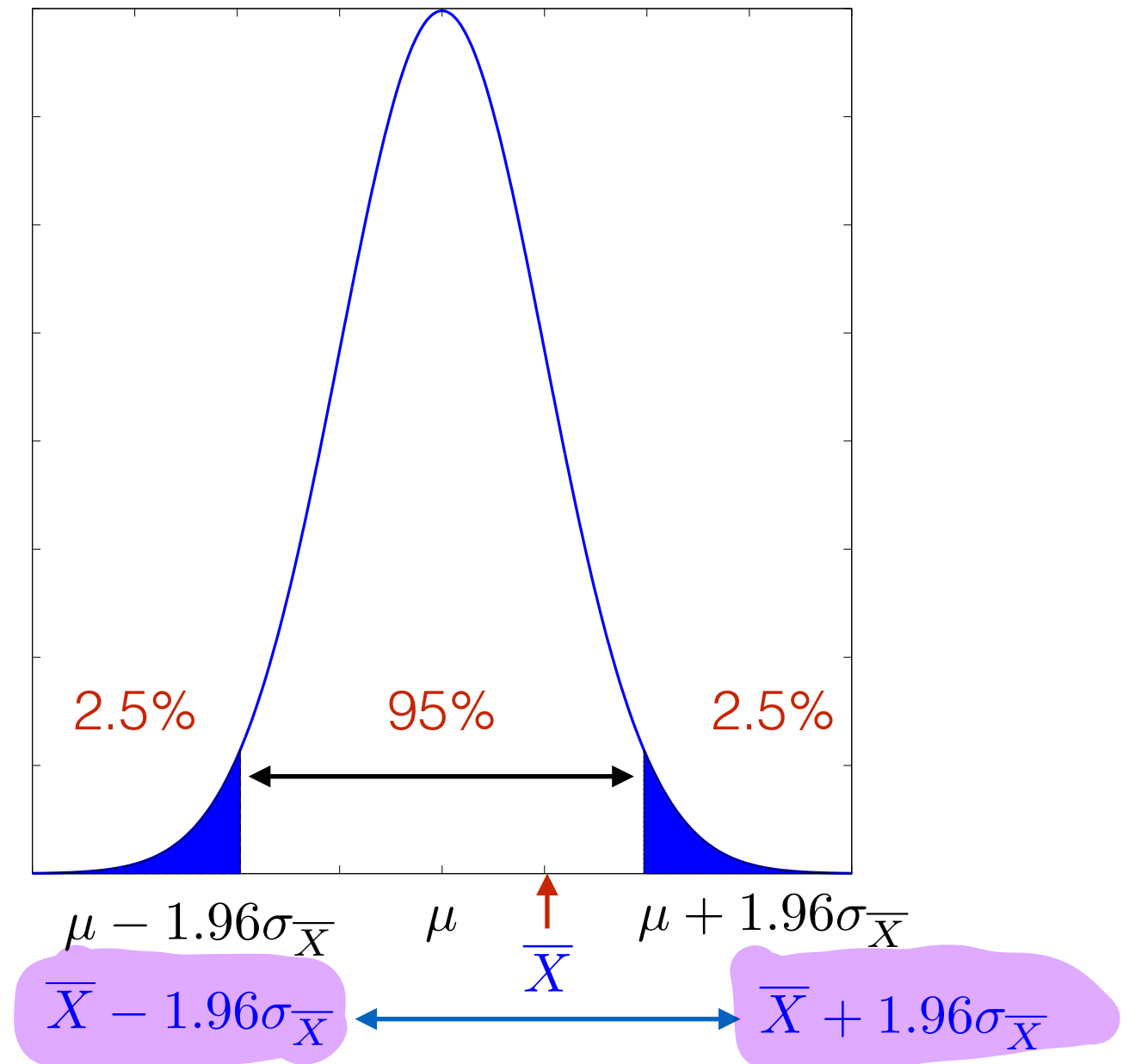
$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

- The middle 95% is within

$$\mu \pm 1.96\sigma_{\bar{X}}$$

95% confidence interval

$$\bar{X} \pm 1.96\sigma_{\bar{X}}$$

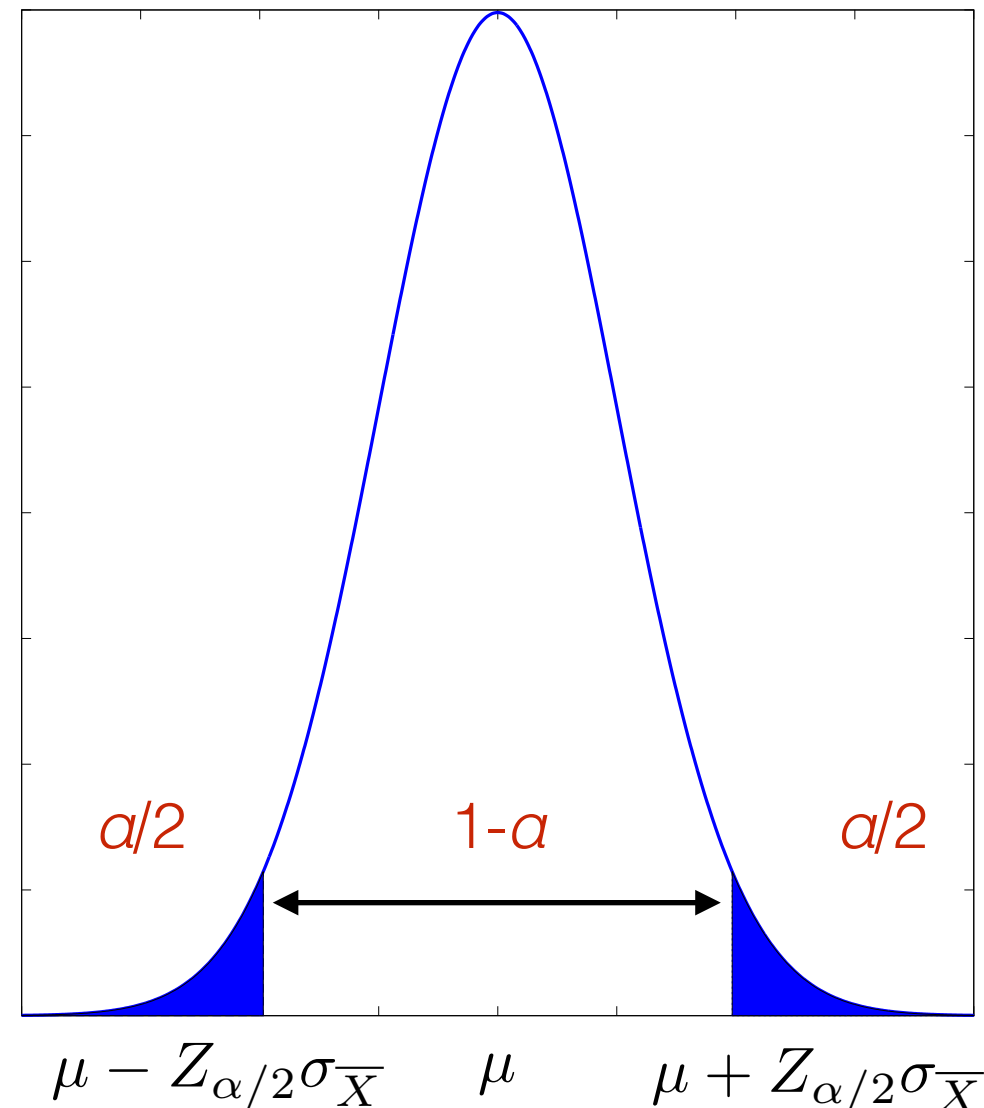




# Confidence Interval

- Let  $X_1, X_2, \dots, X_n$  be a large ( $n \geq 30$ ) random sample from a population with mean  $\mu$  and standard deviation  $\sigma$ .
- Let  $\alpha$  be a number in  $[0,1]$
- Let  $z_{\alpha/2}$  be the z-score that cuts off an area of  $\alpha/2$  in the right-tail
- 100(1- $\alpha$ )% confidence interval for  $\mu$  is

$$\bar{X} \pm Z_{\alpha/2} \sigma_{\bar{X}}$$





# Large-Sample Confidence Intervals for a Population Mean

- Let  $X_1, X_2, \dots, X_n$  be a large ( $n \geq 30$ ) random sample from a population with mean  $\mu$  and standard deviation  $\sigma$ , so that the sample mean  $\bar{X}$  is approximately normal. Then a level  $100(1 - \alpha)\%$  confidence interval for  $\mu$  is

$$\bar{X} \pm Z_{\alpha/2} \sigma_{\bar{X}}$$

where  $Z_{\alpha/2}$  is the z-score that cuts off an area of  $\alpha/2$  on the right-tail

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

When the value of  $\sigma$  is unknown, it can be replaced with sample standard deviation  $s$ .

# Values of $Z_{\alpha/2}$ at different C.I.

| C.I. Level | $Z_{\alpha/2}$ |
|------------|----------------|
| 68%        | 1              |
| 90%        | 1.645          |
| 95%        | 1.96           |
| 99%        | 2.58           |
| 99.7%      | 3              |

# Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are  $\bar{X} = 12.05$  and  $s = 0.1$ . Find
  - 68% confidence interval

$$\alpha = 0.32, \alpha/2 = 0.16$$

$$Z_{\alpha/2} = 1 \quad (1 - 0.8413 \approx 0.16)$$

$$\bar{X} \pm Z_{\alpha/2} s \sqrt{n}$$

$$12.05 \pm 1 \cdot \frac{0.1}{\sqrt{100}}$$

$$68\% \text{ CI is } (12.04, 12.06)$$

0.32

# Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are  $\bar{X} = 12.05$  and  $s = 0.1$ . Find

- 85% confidence interval

$$\alpha = 0.15, \alpha/2 = 0.075$$

$$Z_{\alpha/2} = 1.44 \quad (\pm 0.9251 \approx 0.075)$$

$$\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$12.05 \pm 1.44 \times \frac{0.1}{\sqrt{100}}$$

$$85\% \text{ is } (12.036, 12.064)$$

# Example 3

- A sample of 50 students has the following sample mean and standard deviation for heights,  $\bar{X} = 164.2$  cm and  $s = 12.5$  cm. Find
  - 68% confidence interval
  - 95% confidence interval
  - 99% confidence interval

# Example 4

- Find the levels of the confidence intervals that have the following values of  $Z_{\alpha/2}$ 
  - $Z_{\alpha/2} = 1.28$
  - $Z_{\alpha/2} = 2.17$
  - $Z_{\alpha/2} = 3.28$

# Determining Sample Size Needed for a C.I. with Specified Width

- Sometimes the C.I. is too wide to be useful, and we want to make it narrower. To do this, the sample size must be increased.
- Ex: The sample standard deviation of weights from 100 boxes was  $s = 0.1$  oz. How many boxes must be sampled to obtain a 99% confidence interval of width  $\pm 0.012$  oz

$$\alpha = 0.01, \frac{\alpha}{2} = 0.005$$

$$Z_{0.005} = 2.58$$

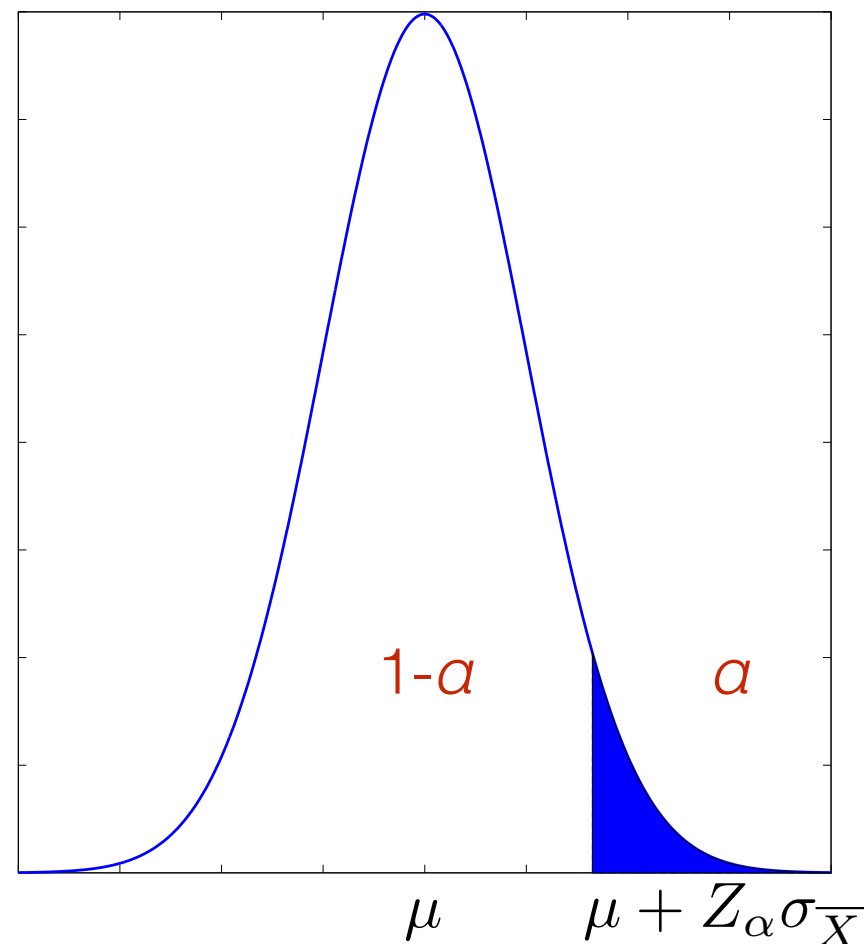
$$0.012 = 2.58 \times \frac{0.1}{\sqrt{n}}$$

$$n \approx 463$$



# One-sided C.I.

- The confidence intervals discussed so far have been two-sided in that they specify both a lower and upper confidence bounds
- In some applications, we are only interested only in one of these bounds





# One-sided C.I. (2)

- Let  $X_1, X_2, \dots, X_n$  be a large ( $n \geq 30$ ) random sample from a population with mean  $\mu$  and standard deviation  $\sigma$ , so that the sample mean  $\bar{X}$  is approximately normal. Then a level  $100(1 - \alpha)\%$  lower confidence interval bound for  $\mu$  is

$$\bar{X} - Z_{\alpha} \sigma_{\bar{X}}$$

and level  $100(1 - \alpha)\%$  upper confidence bound or  $\mu$  is

$$\bar{X} + Z_{\alpha} \sigma_{\bar{X}}$$

When the value of  $\sigma$  is unknown, it can be replaced with sample standard deviation  $s$ .

# Values of $Z_\alpha$ at different C.I. (One-sided)

| C.I. Level | $Z_\alpha$ |
|------------|------------|
| 90%        | 1.28       |
| 95%        | 1.645      |
| 99%        | 2.33       |

# Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights,  $\bar{x} = 164.2$  cm and  $s = 12.5$  cm. Find
- one-sided 95% lower confidence bound

$$\bar{x} - z_{\alpha} \sigma_{\bar{x}}$$

$$\alpha = 0.05$$

$$z_{0.05} = 1.645$$

$$\bar{x} - z_{\alpha} \sigma_{\bar{x}}$$

$$164.2 - 1.645 \times \frac{12.5}{\sqrt{50}} = 161.29$$

# Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights,  $\bar{x} = 164.2$  cm and  $s = 12.5$  cm. Find
  - one-sided 99% upper confidence bound

$$\bar{x} + z_{\alpha} \frac{s}{\sqrt{n}} \quad \alpha = 0.01$$

$$164.2 + 2.33 \times \frac{12.5}{\sqrt{50}} = 168.32$$